H&M Personalized Fashion Recommendations: Data Dictionary

1. Overview

Item	Description
Source	Kaggle, H&M Personalized Fashion Recommendations competition
Content	Purchase history of H&M customers, plus metadata for customers and articles
Period	About 2 years of transactions (Sept 2018 to Sept 2020)
Task	Predict which articles each customer will buy in the 7 days after the training period ends
Granularity	`transactions_train`: one row per purchased item. `customers`: one row per customer. `articles`: one row per article

Files

File	Purpose
`transactions_train.csv`	Purchases (the main table)
`customers.csv`	Customer metadata
`articles.csv`	Article (product) metadata
`images/`	Article images, stored in subfolders named by the first 3 digits of `article_id`. Not every article has an image
`sample_submission.csv`	Example of the submission format

Relationships

```
customers (1) ———< transactions_train >——— (1) articles
 customer_id                                  article_id
```

Each customer can have many transactions, and each article can appear in many transactions.

2. Transactions (`transactions_train.csv`)

Column	Type	Description	Notes
`t_dat`	datetime	Date of purchase	Loaded as a string, so convert with `pd.to_datetime`. Used for

			time-based train/validation splits
customer_id	string	Customer who bought the item	Foreign key to customers. Anonymized hash
article_id	int	Article that was bought	Foreign key to articles
price	float	Price paid	Normalized, not the real currency value. Only compare it relative to other prices
sales_channel_id	int	Where the purchase happened	1 = in store, 2 = online

Duplicate rows are not errors. They mean the same customer bought the same article multiple times on the same day. Don't drop them blindly.

3. Customers (customers.csv)

Column	Type	Description	Notes
customer_id	string	Unique customer identifier	Primary key. Anonymized hash
FN	float	Flag for whether the customer gets the fashion news newsletter	1.0 = yes, missing = no. Most values are missing
Active	float	Flag for whether the customer is active for communication	1.0 = yes, missing = no. Most values are missing
club_member_status	category	Membership status in the H&M club	ACTIVE, PRE-CREATE, LEFT CLUB
fashion_news_frequency	category	How often the customer receives fashion news	NONE, Regularly, Monthly
age	float	Customer age in years	Has some missing values, and outliers worth checking
postal_code	string	Customer's postal code	Anonymized hash, so it is only useful for grouping customers

For FN and Active, missing means "no", so fill with 0 rather than treating it as unknown.

4. Articles (`articles.csv`)

Many columns come as a **code/number + name** pair that describe the same thing. Use the number for modeling and the name for reading and EDA.

Identification

Column	Type	Description
`article_id`	int	Unique ID of one specific article (a product in a specific colour). Primary key
`product_code`	int	Code of the product itself. One product can have several articles (e.g. the same shirt in different colours)
`prod_name`	string	Name of the product

Product type

Column	Description	Example
`product_type_no` / `product_type_name`	Specific type of product	Trousers, Dress, Sweater
`product_group_name`	Broader group of product types	Garment Upper body, Accessories

Appearance

Column	Description	Example
`graphical_appearance_no` / `graphical_appearance_name`	Pattern or graphic style	Solid, Stripe, All over pattern
`colour_group_code` / `colour_group_name`	Specific colour	Black, Dark Blue
`perceived_colour_value_id` / `perceived_colour_value_name`	How light or dark the colour looks	Dark, Light, Dusty Light
`perceived_colour_master_id` / `perceived_colour_master_name`	Base colour family	Blue, Black, Beige

Store organization (from specific to broad)

Column	Description	Example
`department_no` / `department_name`	Department that handles the article	Jersey Basic, Trousers
`section_no` / `section_name`	Section in the store	Womens Everyday Basics
`garment_group_no` / `garment_group_name`	Group of similar garments	Jersey Fancy, Knitwear

index_code / index_name	Main category index, shown as a letter code	Ladieswear, Menswear, Sportswear
index_group_no / index_group_name	Highest-level category	Ladieswear, Baby/Children, Menswear, Sport, Divided

Text

Column	Type	Description
detail_desc	string	Free-text description of the article. Has a few missing values. Useful for NLP or text embeddings

5. Notes

- article_id has 10 digits. The first 7 are the product_code, and the last 3 are the colour variant.
- To find an image, use images/<first 3 digits of article_id>/<article_id>.jpg.
- Check missing values and exact unique values yourself with df.isna().sum() and df[col].value_counts(). I wrote the category values from my knowledge of the dataset, and your version may differ slightly.